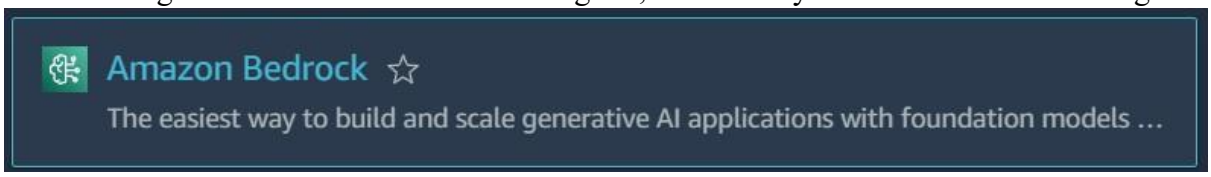
Amazon Bedrock (Image Generation)

To Begin with the Lab

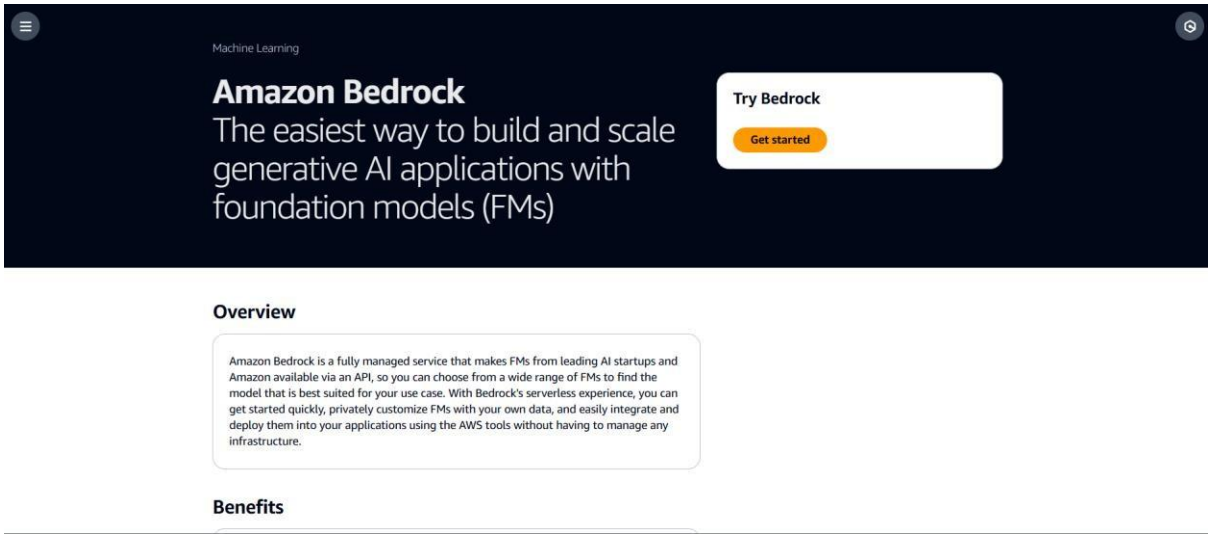
Summary of the Lab

In your AWS Console search for Amazon Bedrock. Choose this service accordingly.

The region used in this lab is North Virginia, make sure you are also in the same region.

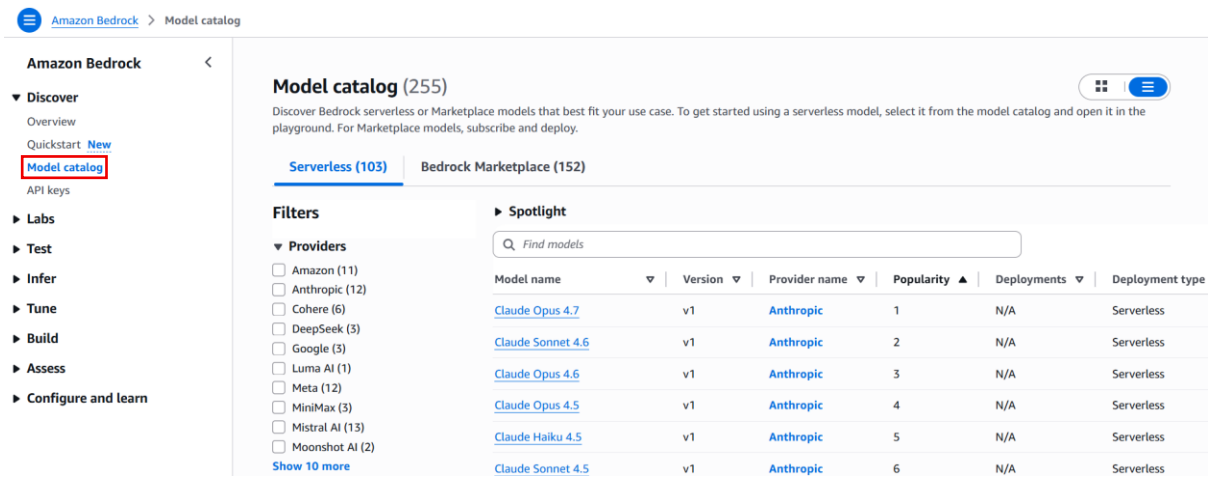


This is the landing page Amazon Bedrock.



Go to Bedrock → Model Catalog and request access for models (some are instant, others require approval).

Bedrock is available across multiple regions, but not all models are in every region—so choose a region where your model is supported.

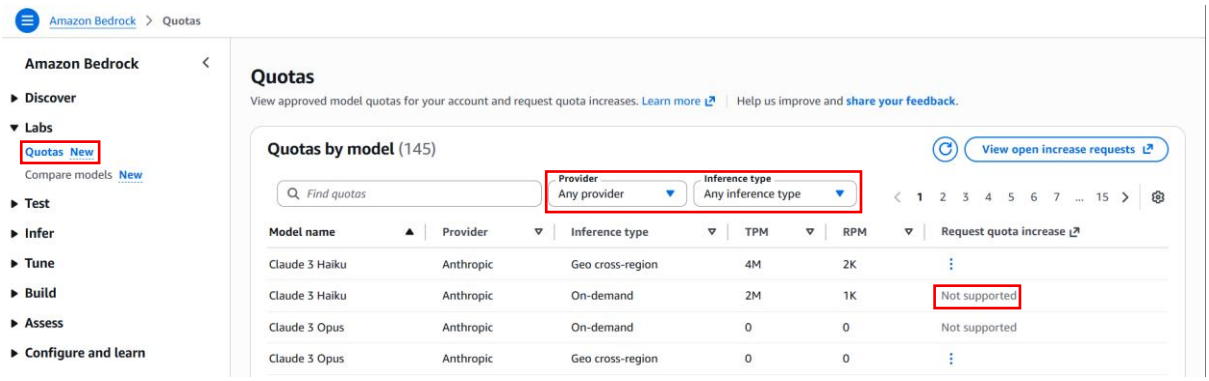


Note: To Check which models are available in your region for you.

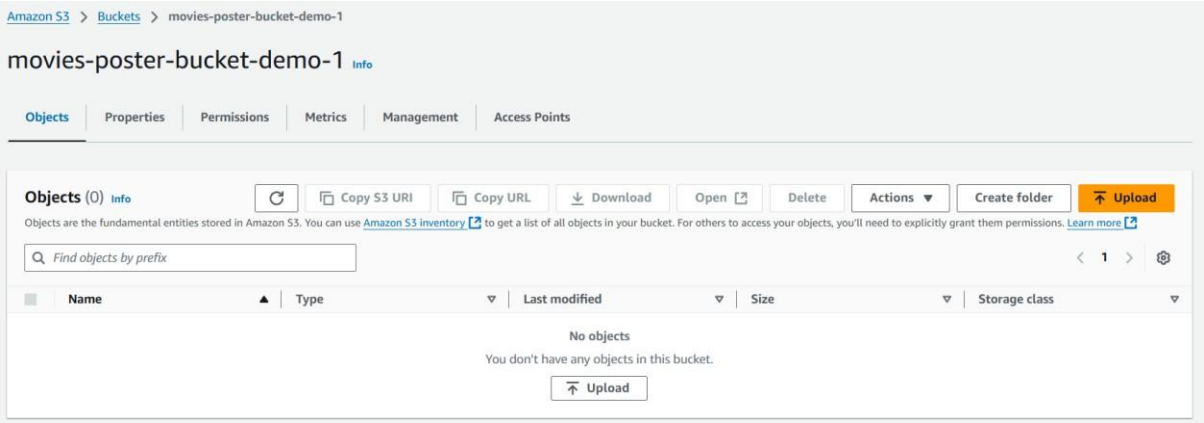
Go to Labs → Quotas

According to your need, you can select Provider and Inference Type

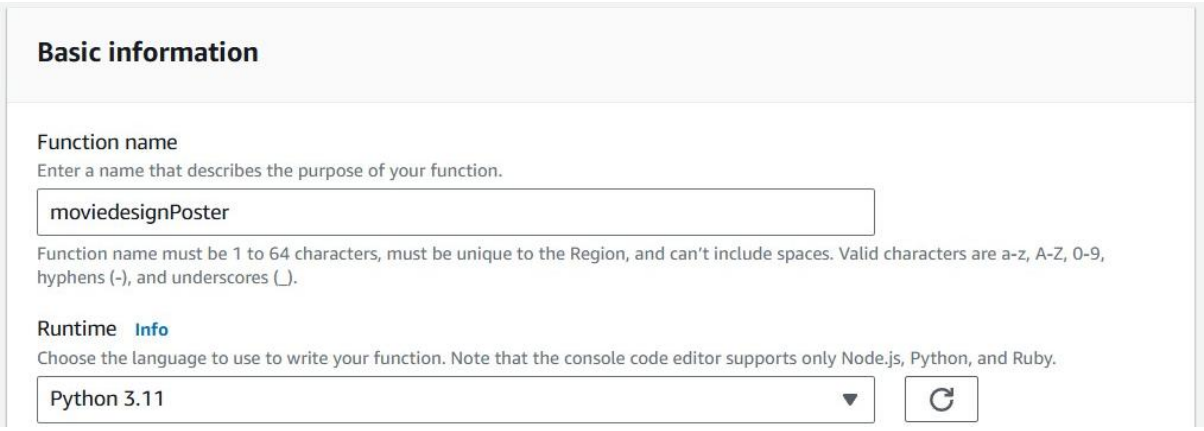
You can see the models which are supported and Not supports



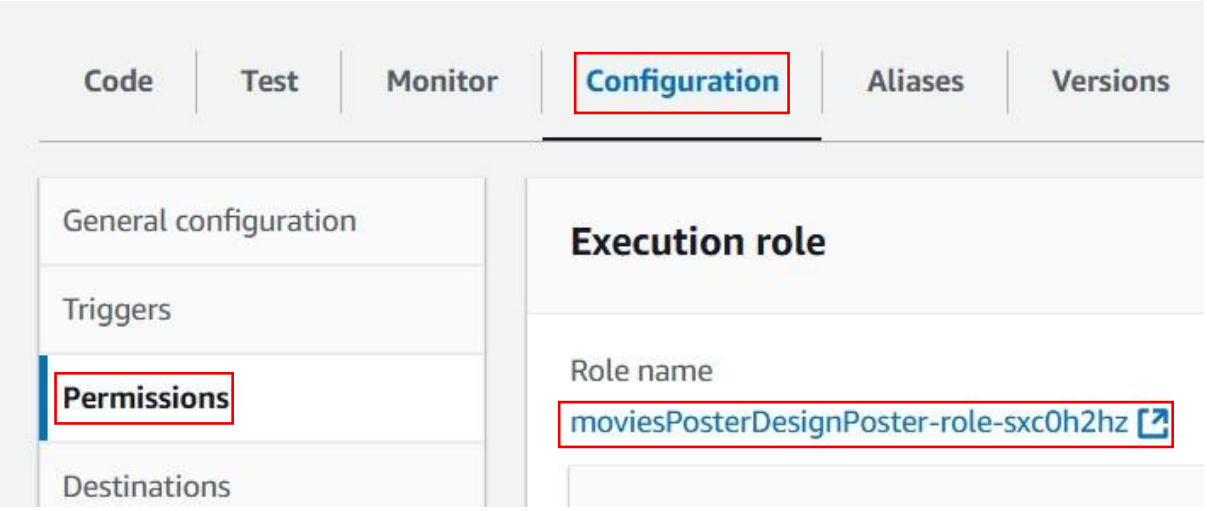
After all that first we need to search S3 and create a bucket. Below you can see that our bucket has been created successfully.



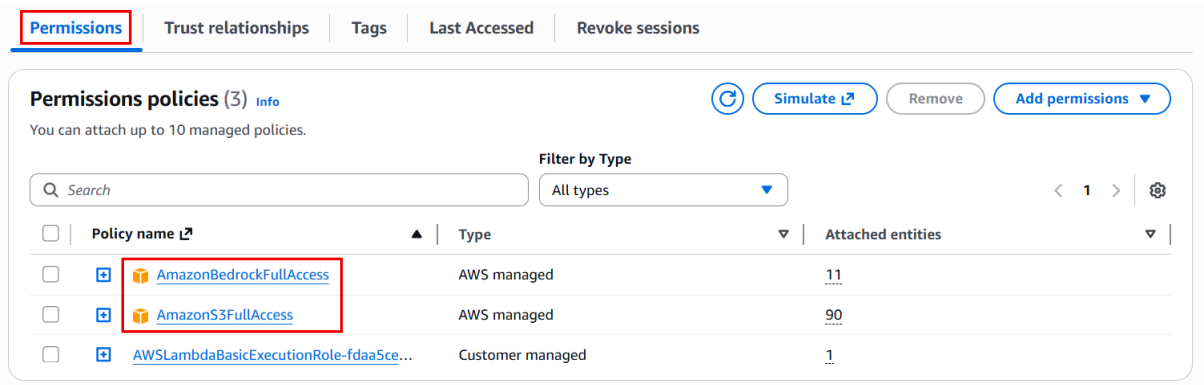
Then we need to create a Lambda function, search and navigate to lambda. Here you just need to choose Author from scratch and give it a name then choose Python 3.11 as your environment and keep rest of the settings to default and create your lambda function.



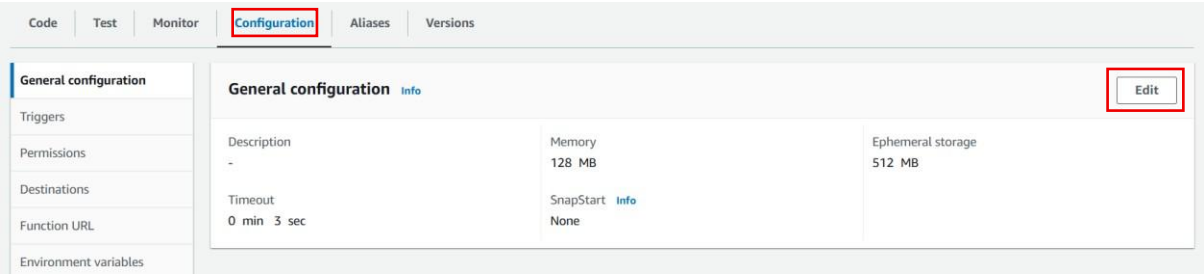
Since your lambda function has been created, you need to change some of its properties. For that, you need to go to the configuration tab, then to Permission, and click on the IAM role attached to your lambda function.

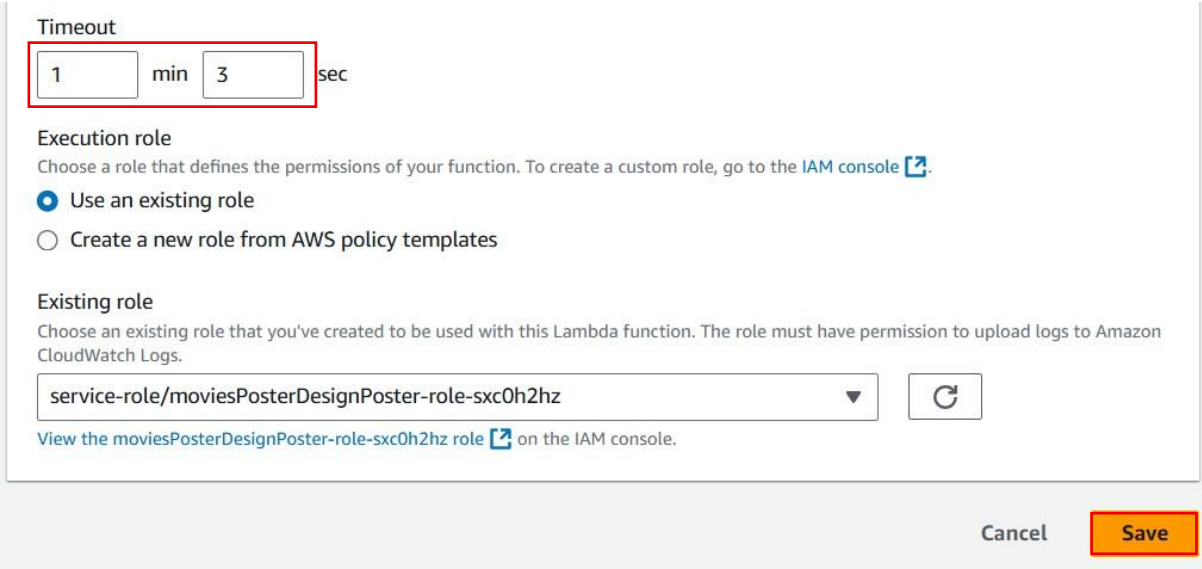


To this IAM role, for the simplicity of this lab attach required permission to your IAM role and come back to lambda function. Now IAM role has access to everything.



Then come to the general configuration and click on edit. Here we are going to increase the timeout limit to 1 minute from just 3 seconds. Click on save to save your changes.





After that you need to come back to the code tab and paste the code shown below. You will also get a separate file for this code. Then click on deploy to deploy your code

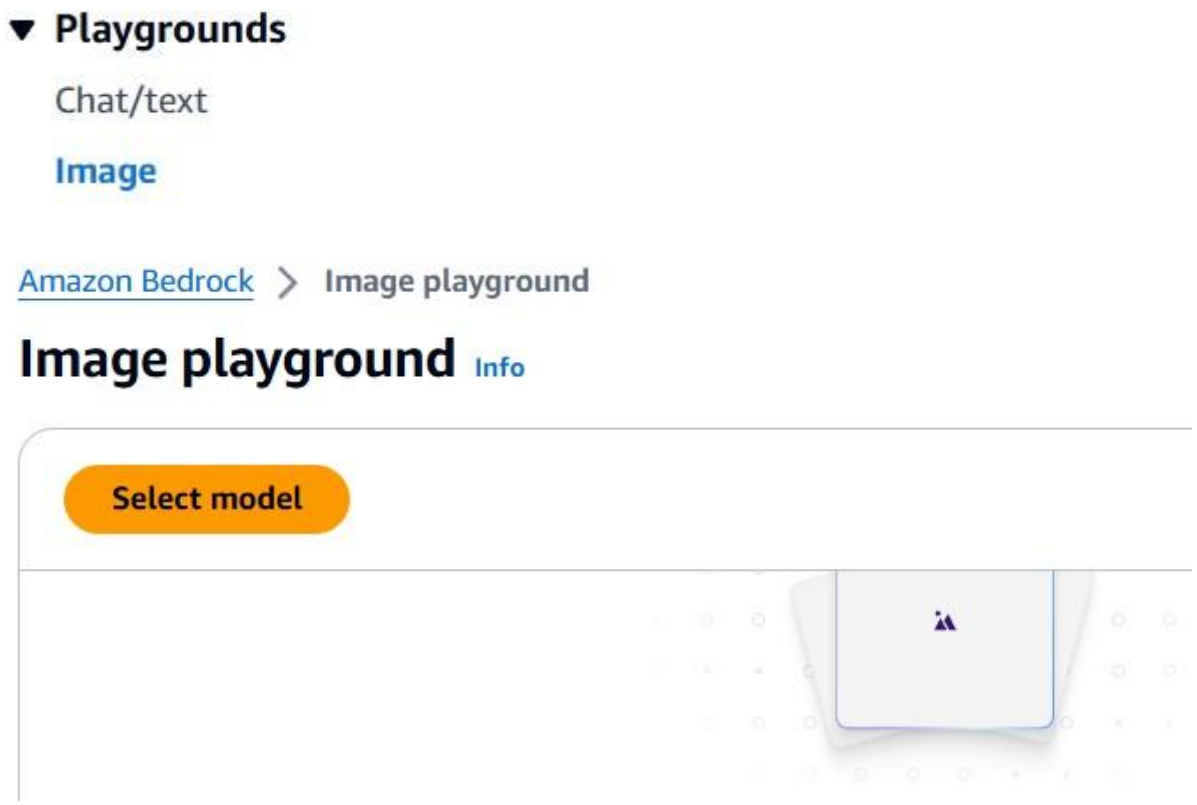
```
import json

import boto3

import base64

from datetime import datetime


bedrock_client = boto3.client("bedrock-runtime")

s3_client = boto3.client("s3")


BUCKET_NAME = "amzn-web-infrat2"

MODEL_ID = "stability.sd3-5-large-v1:0"



def lambda_handler(event, context):

try:

prompt = event.get("prompt", "A dog near the sea at sunset")
```

```
request_body = {
"prompt": prompt,
"seed": 0,
"output_format": "png",
"aspect_ratio": "1:1"
}

response = bedrock_client.invoke_model(
modelId=MODEL_ID,
contentType="application/json",
accept="application/json",
body=json.dumps(request_body)
)

response_body = json.loads(response["body"].read())

if "images" not in response_body or not response_body["images"]:
raise ValueError("No image returned from model")

image_base64 = response_body["images"][0]
image_bytes = base64.b64decode(image_base64)

file_name = f"poster-{datetime.utcnow().strftime('%Y%m%d-%H%M%S')}.png"

s3_client.put_object(
Bucket=BUCKET_NAME,
```

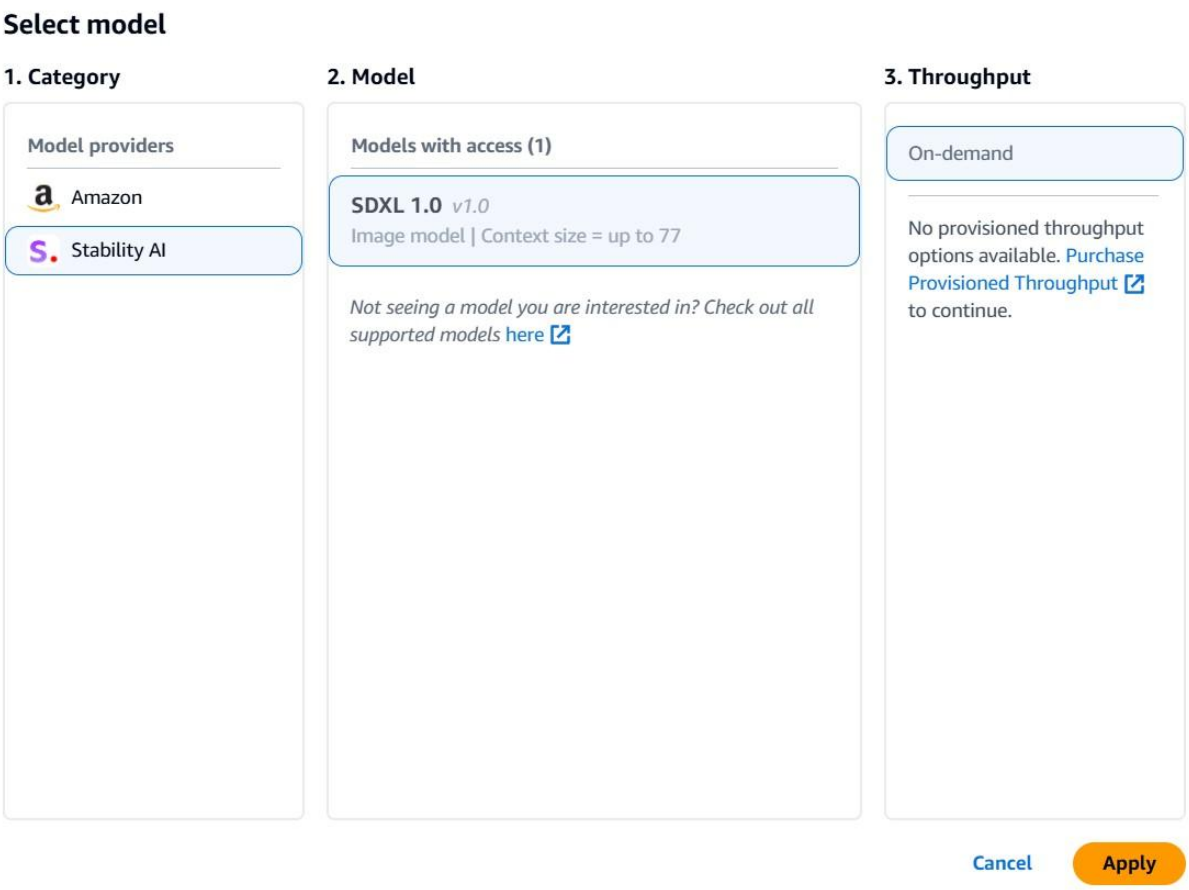
```
Key=file_name,
Body=image_bytes,
ContentType="image/png"
)

presigned_url = s3_client.generate_presigned_url(
"get_object",
Params={"Bucket": BUCKET_NAME, "Key": file_name},
ExpiresIn=3600
)

return {
"statusCode": 200,
"body": json.dumps({
"image_url": presigned_url
})
}

except Exception as e:
return {
"statusCode": 500,
"body": json.dumps({
"error": str(e)
})
}
```

we need a model ID to configure the model to generate an Image and we can get this on the Bedrock console. So, navigate to it.

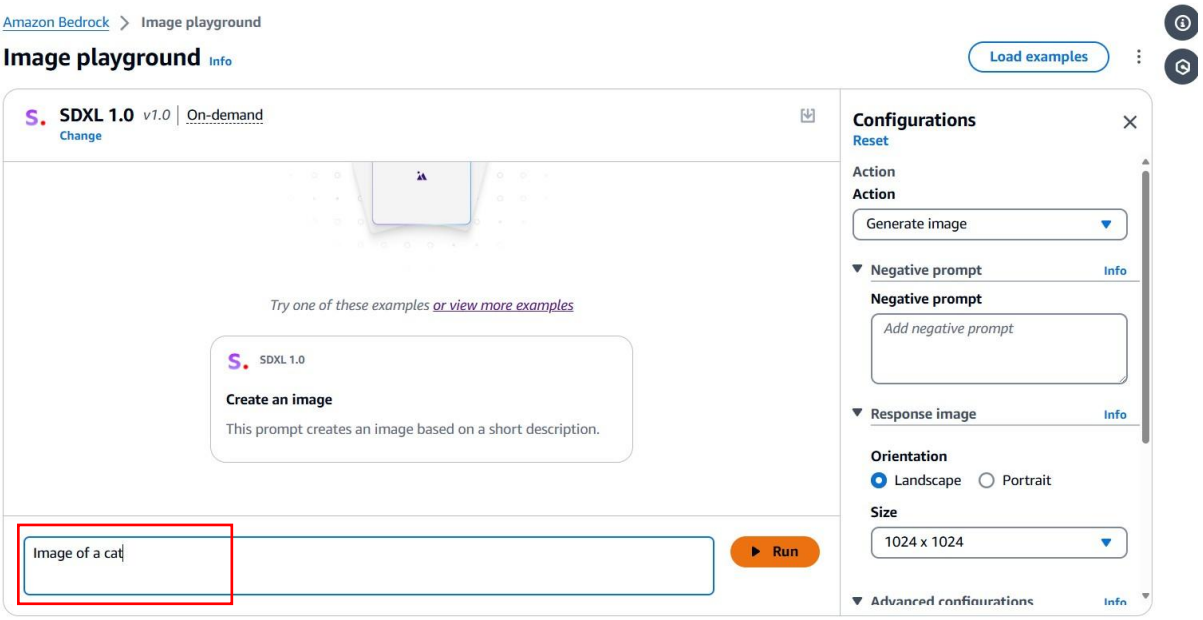
Then from the left pane choose image from Playgrounds. Click on Select model.



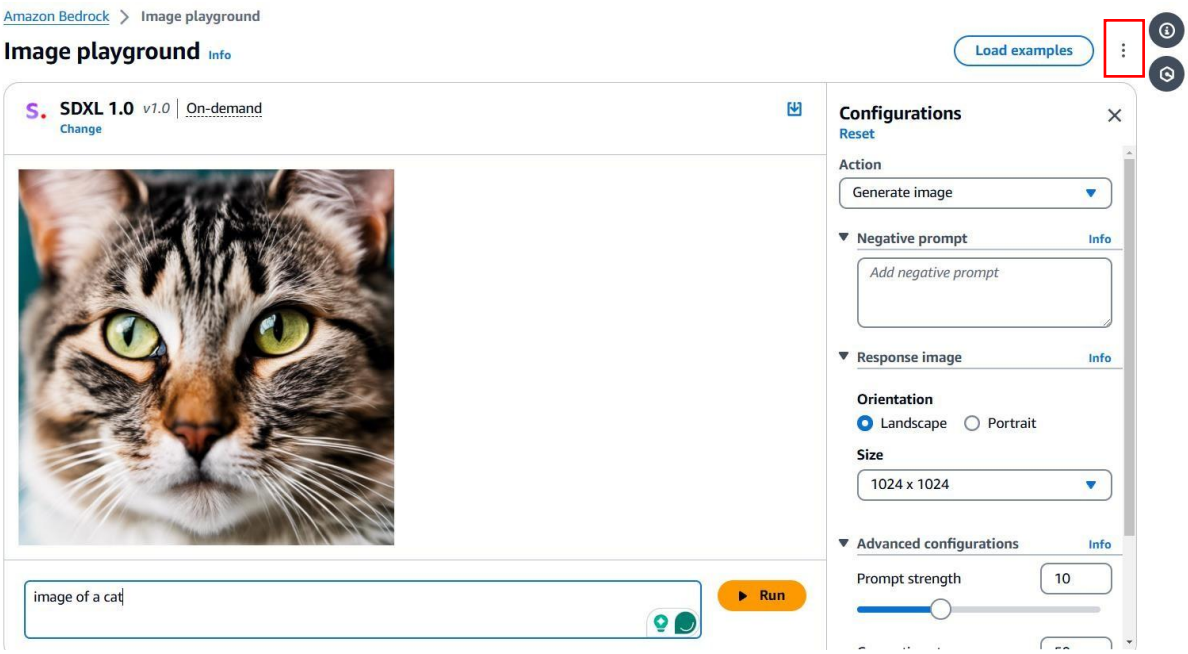
Choose Stability AI and the Model then click on Apply.



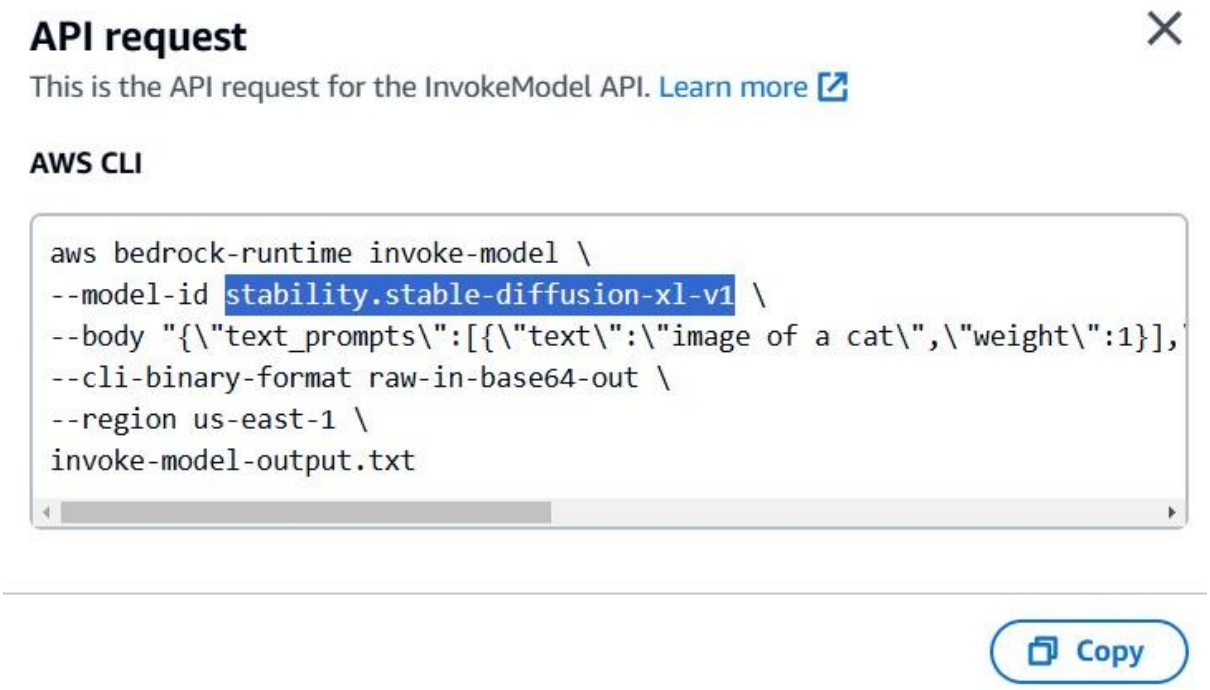
Then in the input text field write a text to generate an image and click on run.



Below you can see that we have an image of a cat generated. Now if you click on the 3 dots highlighted in the snapshot and click on view API request.



From there you need to copy the Model ID and paste it in your lambda function.



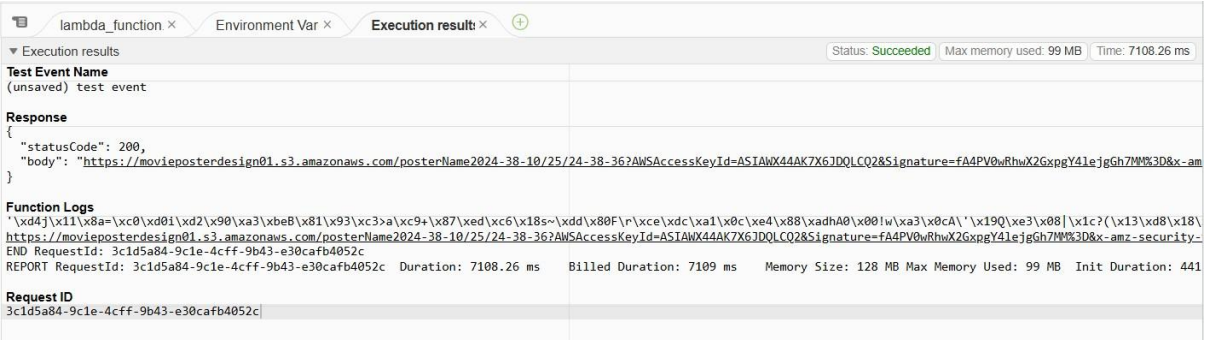
Then in line number 33 and 38 you need to write your bucket name here and click on deploy code.

```
31
32      response_s3=client_s3.put_object(
33          Bucket='movies-poster-design-demo-1',
34          Body=response_bedrock_finalimage,
35          Key=poster_name)
```

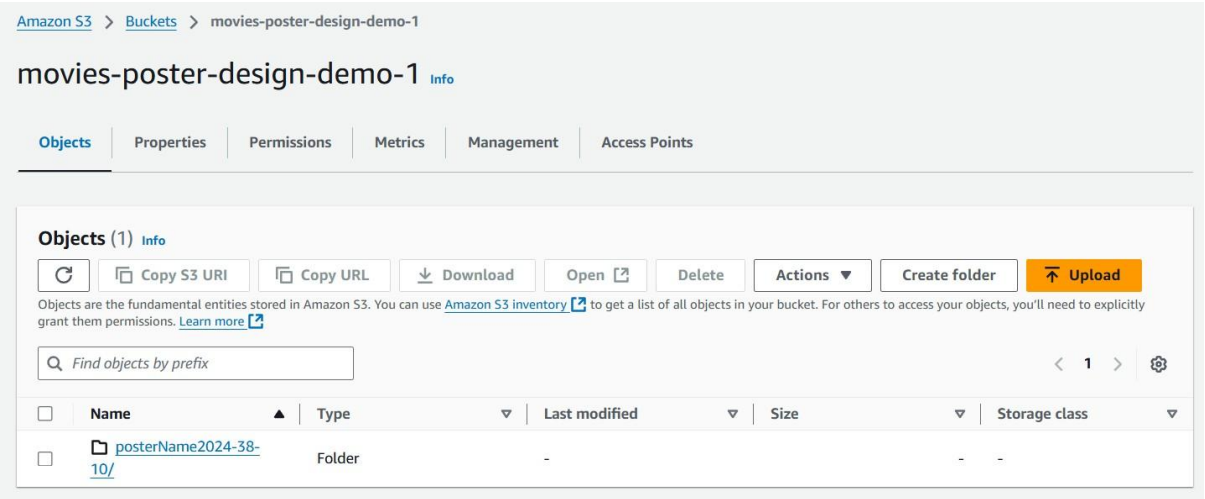
After that you need to create a new test event and in the event JSON area you need to write this. Click on save to save your test then click on test.

{
"prompt": "image of a dog"
}

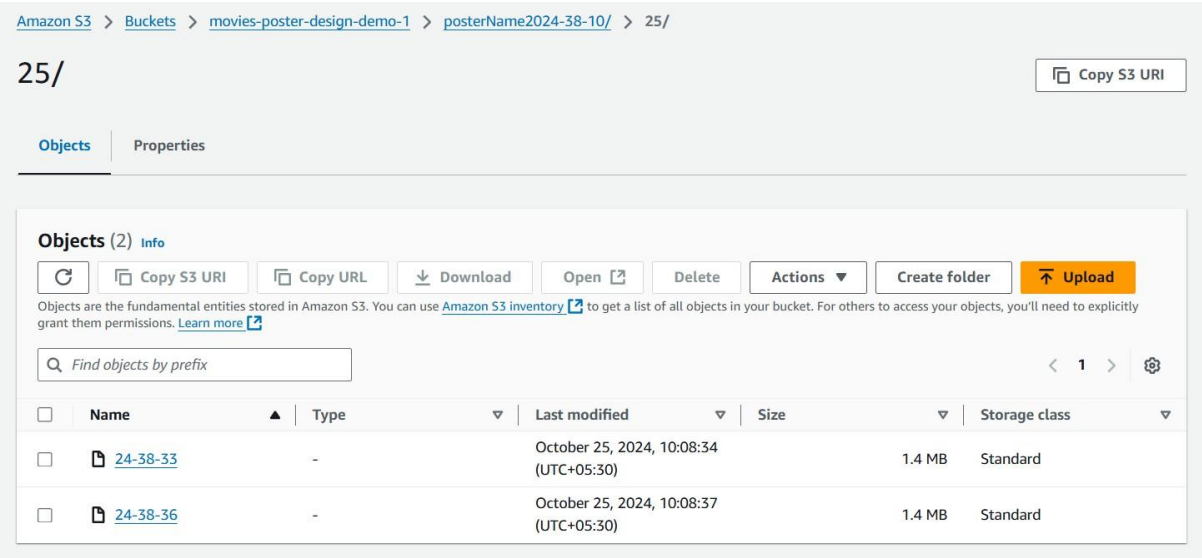
Here you can see that we got status code 200 which means that our image has been generated and it has been stored in our S3 bucket.



Now if go back to my S3 bucket and there I will see a folder.



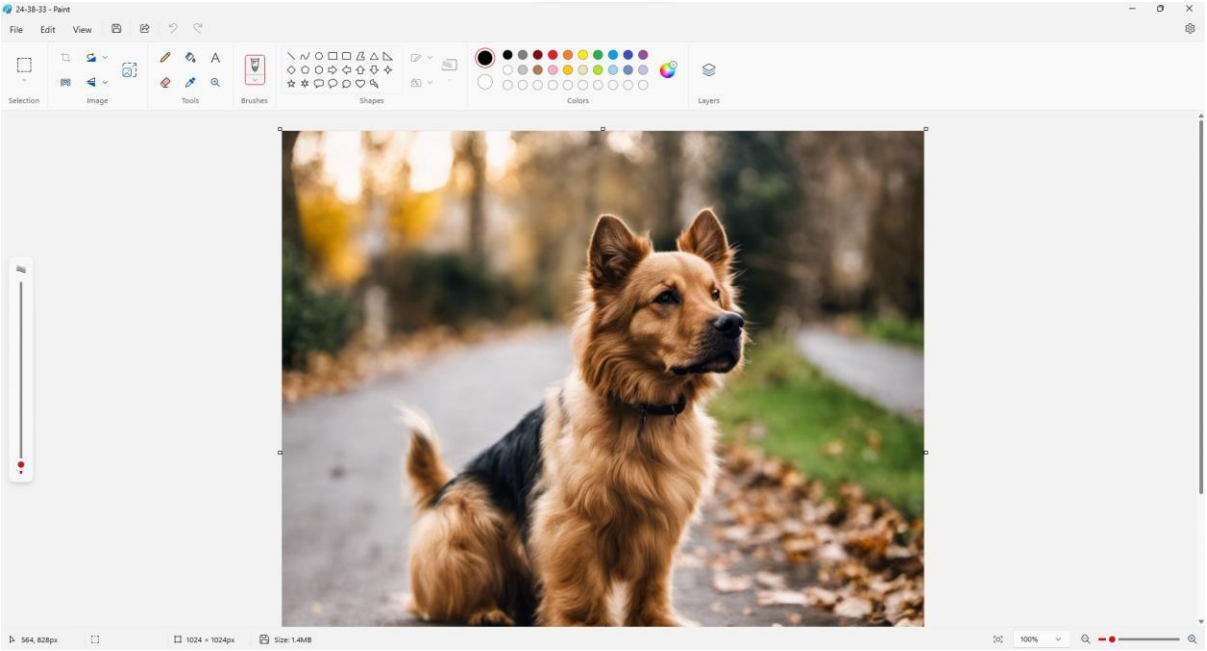
Now if we go inside this folder, we will see some objects download one of the object.



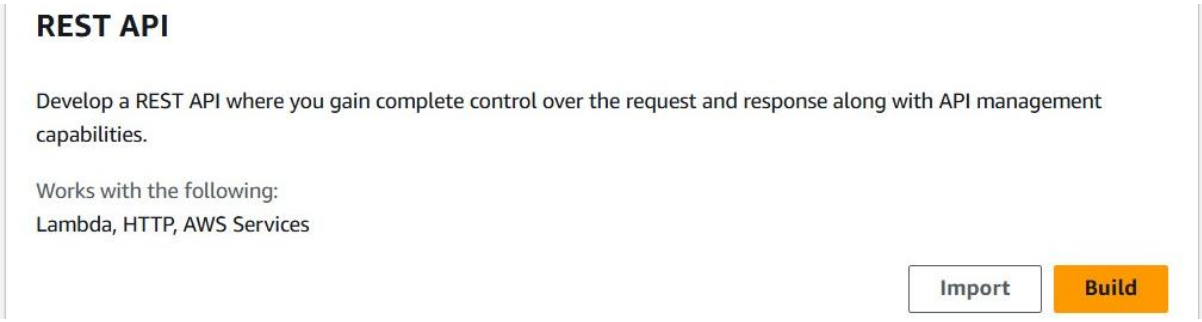
Now what you need to do is, open this downloaded object using Paint on your laptop and you will see an image of a dog.

This was the prompt we had given while saving our test in the lambda function.

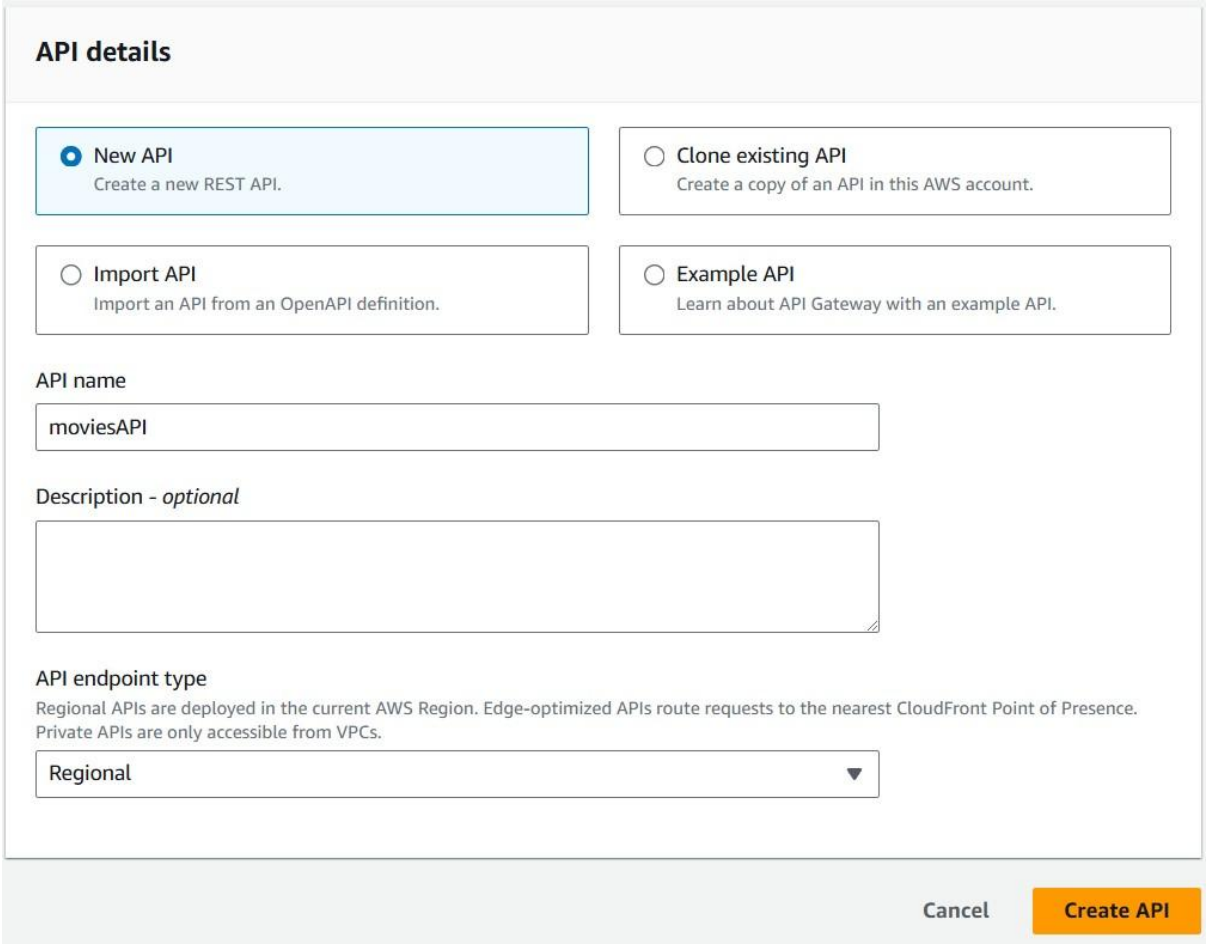
You can also try out a different prompt and test it out using your lambda function.



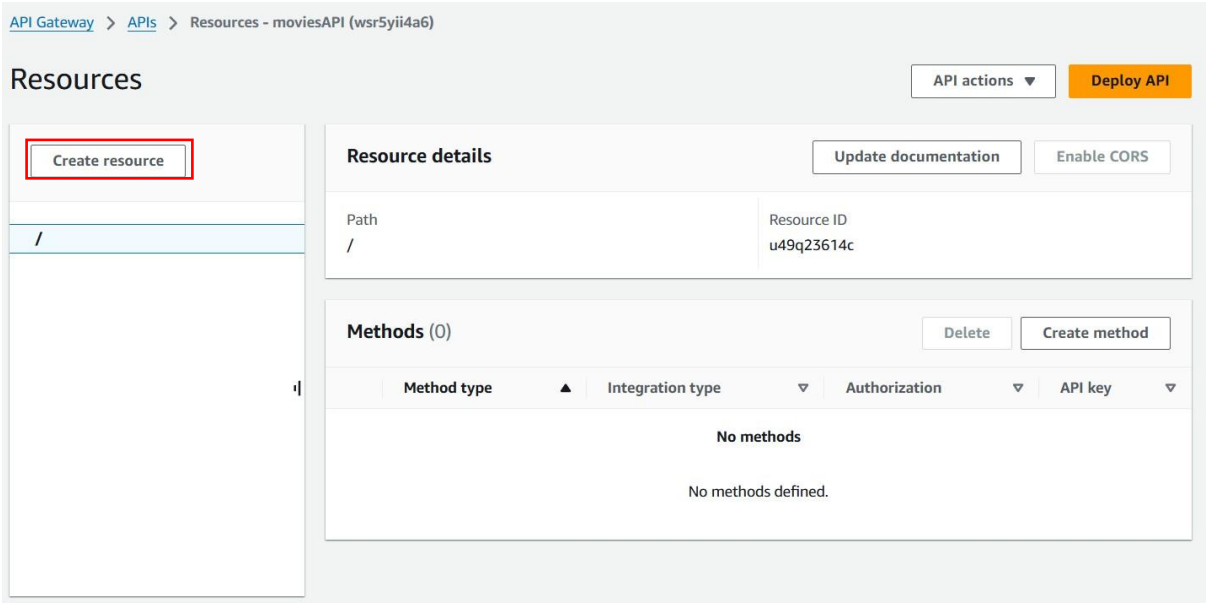
Now you need to search for API gateway in your console and navigate to it. Then you need to click on build.



Here you just need to give it a name and click on Create API.



First, we need to create a resource. Give a name to your resource and enable CORS.



API Gateway > APIs > Resources - moviesAPI (wsr5yii4a6) > Create resource

Create resource

Resource details

Proxy resource Info

Proxy resources handle requests to all sub-resources. To create a proxy resource use a path parameter that ends with a plus sign, for example {proxy+}.

Resource path

/

Resource name

moviesdesign

CORS (Cross Origin Resource Sharing) Info

Create an OPTIONS method that allows all origins, all methods, and several common headers.

Cancel | Create resource

Then you need to select your resource and you need to click on create method.

API Gateway > APIs > Resources - moviesAPI (wsr5yii4a6)

Resources

API actions | Deploy API

Create resource

/

/moviesdesign

OPTIONS

Resource details

Delete | Update documentation | Enable CORS

Path
/moviesdesign

Resource ID
imlpt0

Methods (1)

Delete | Create method

Method type	Integration type	Authorization	API key
OPTIONS	Mock	None	Not required

Choose GET method to integrate it with your lambda function.

Method details

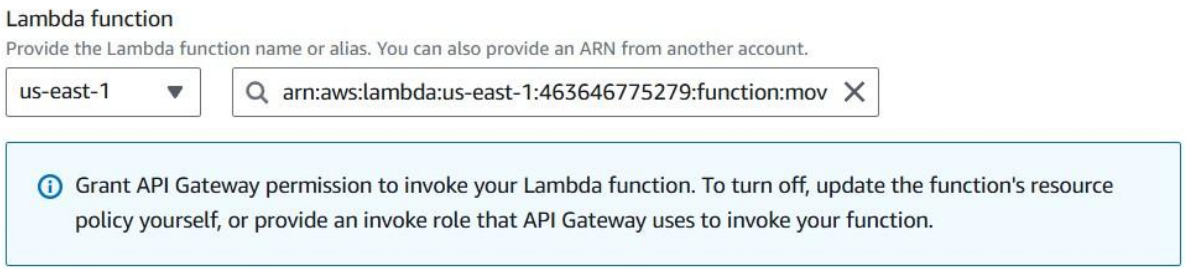
Method type

GET

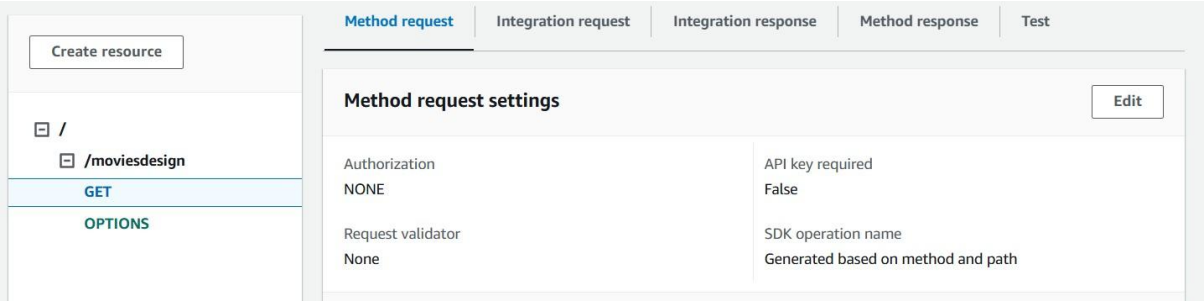
Integration type

- Lambda function
  Integrate your API with a Lambda function.
- HTTP
  Integrate with an existing HTTP endpoint.
- Mock
  Generate a response based on API Gateway mappings and transformations.

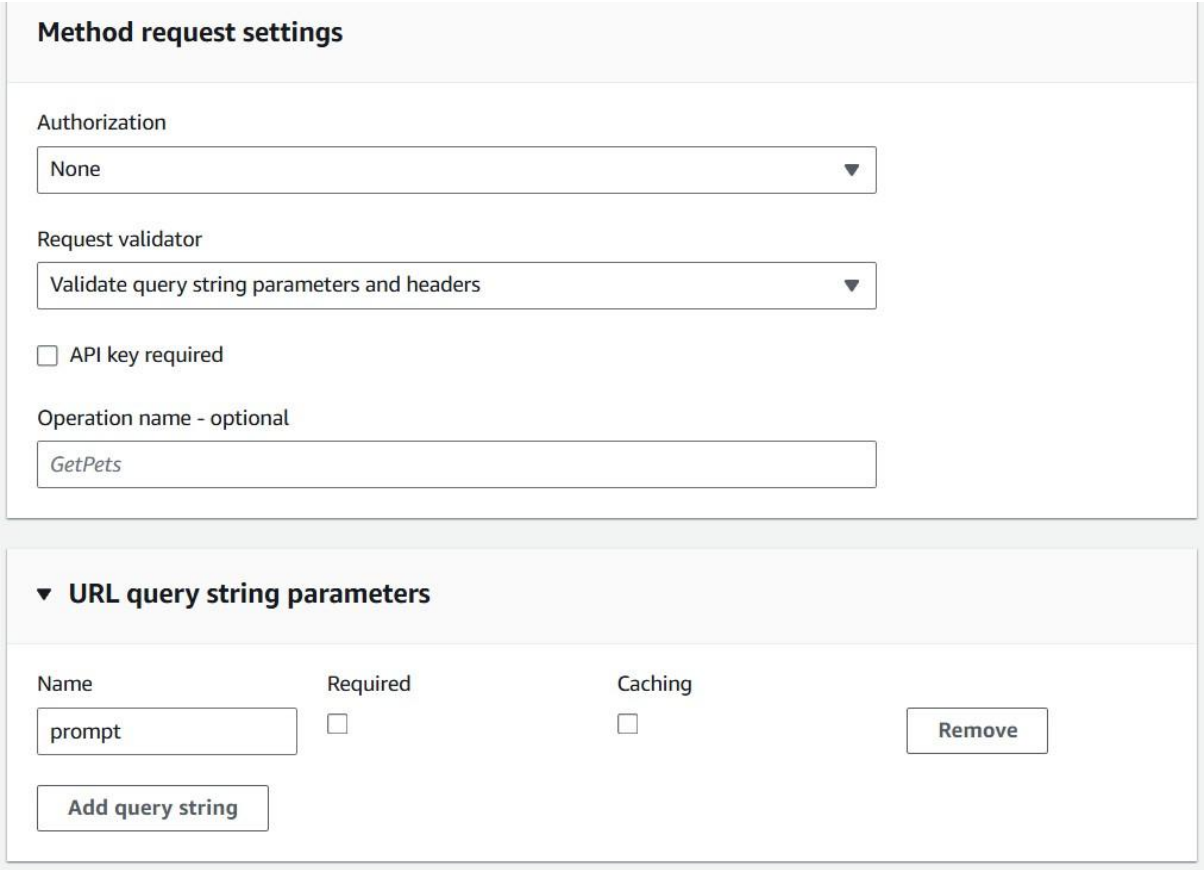
And choose your lambda function and click on create method.



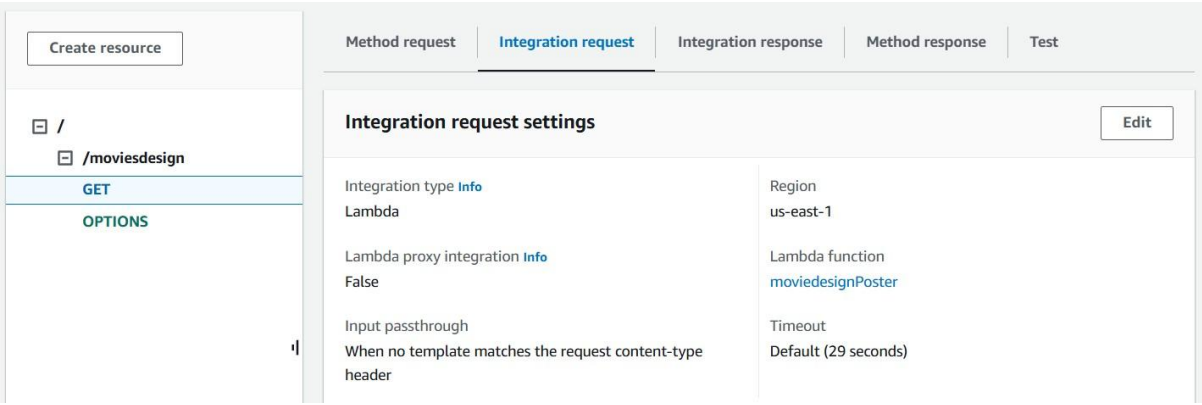
In your GET method go to method request and click on edit.



You need to change the request validator to what shown in the snapshot and expand the URL query string parameters and in the name write prompt as shown below then click on save.

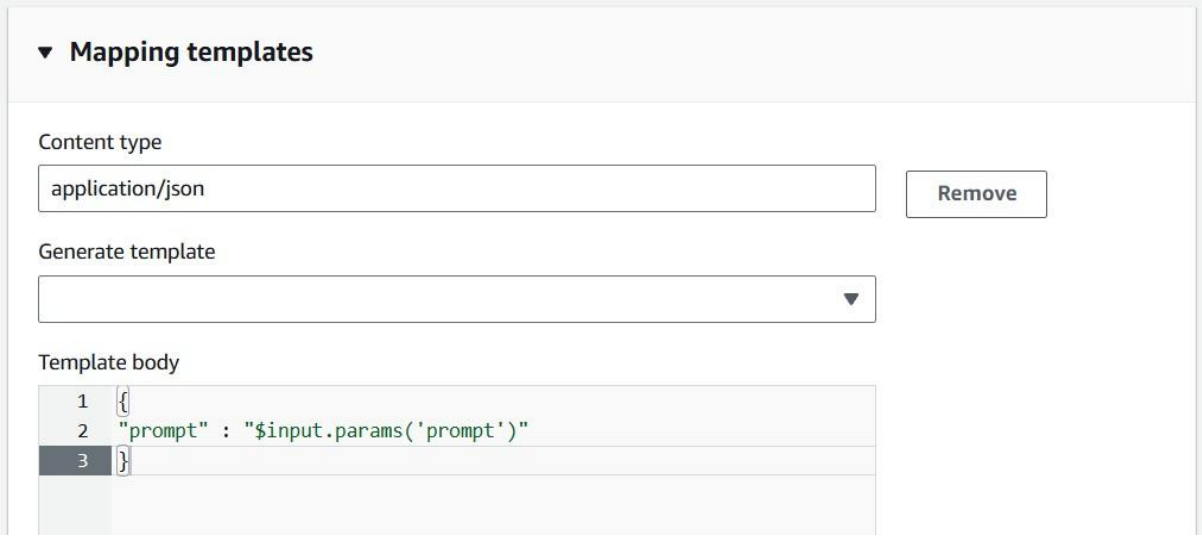


Then go to integration request in your GET method and click on edit.

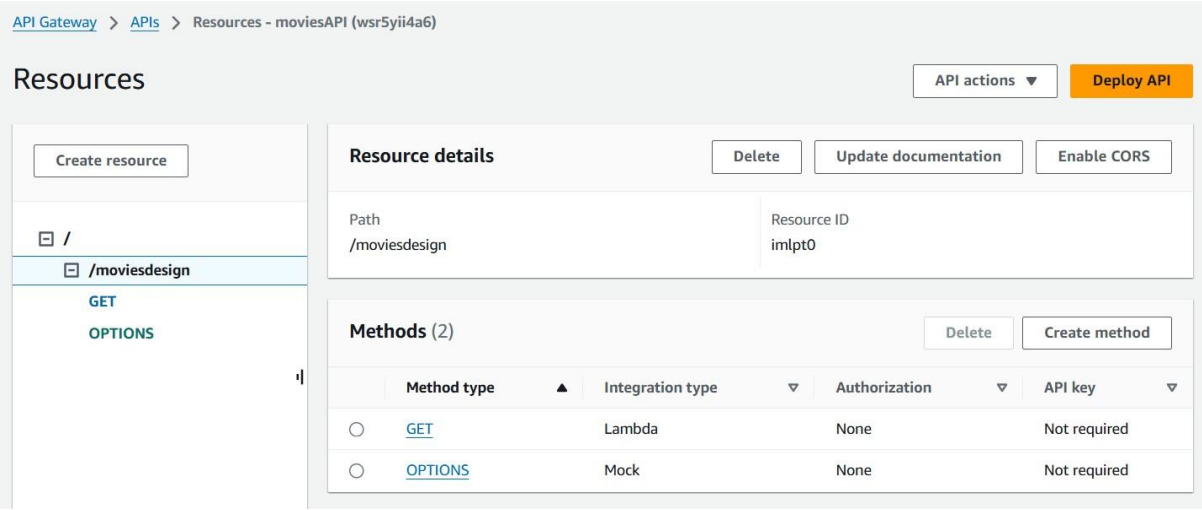


Scroll down to the bottom and expand Mapping templates then choose to add a template. In the content type and template body write the same. Click on save.

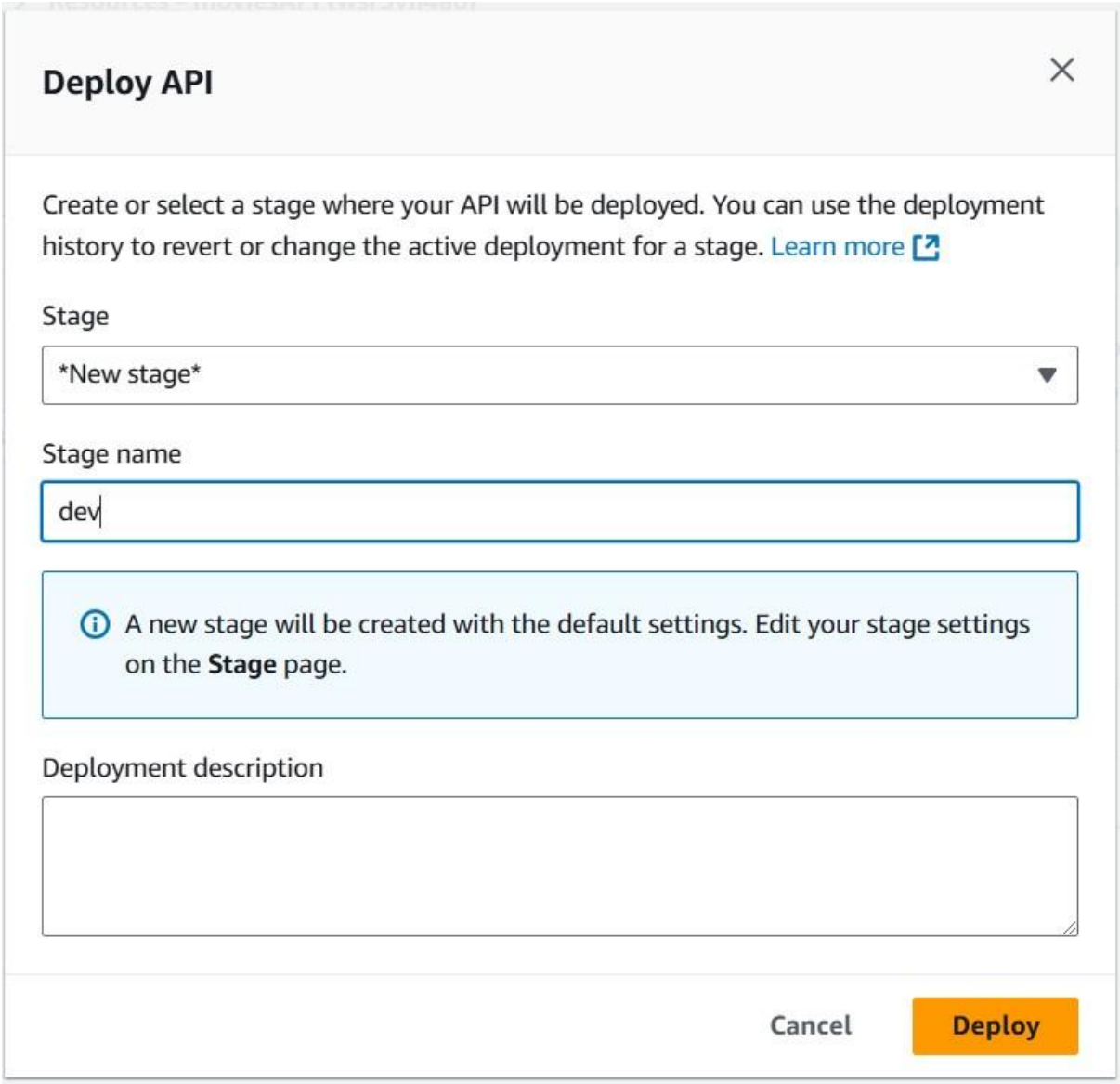
{
"prompt" : "$input.params('prompt')"
}



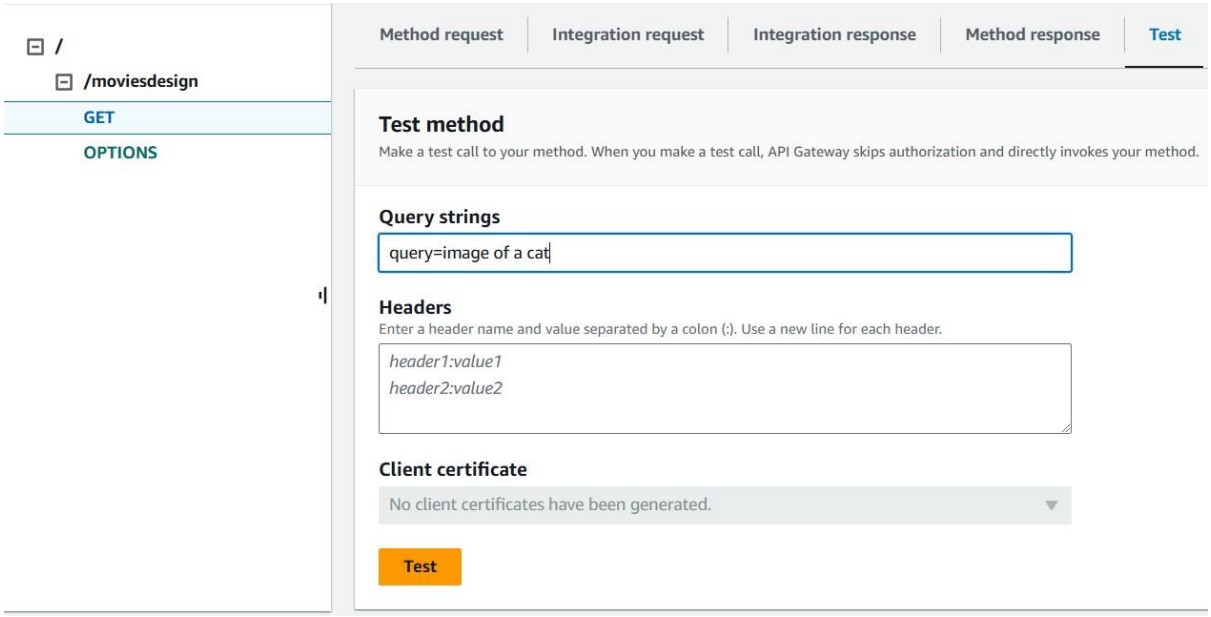
After that you need to deploy your API, for that you need to click on resource and from here choose to deploy you API.



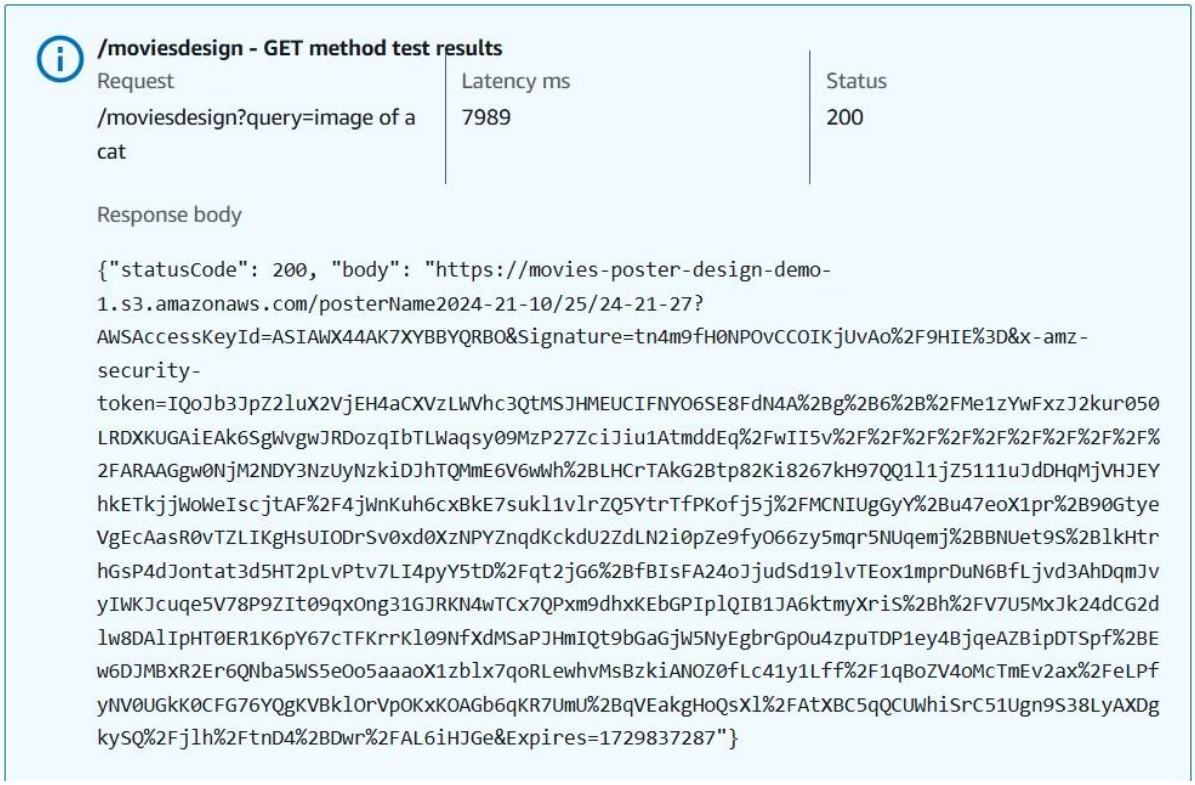
Choose a new stage give it a name and then click on deploy.



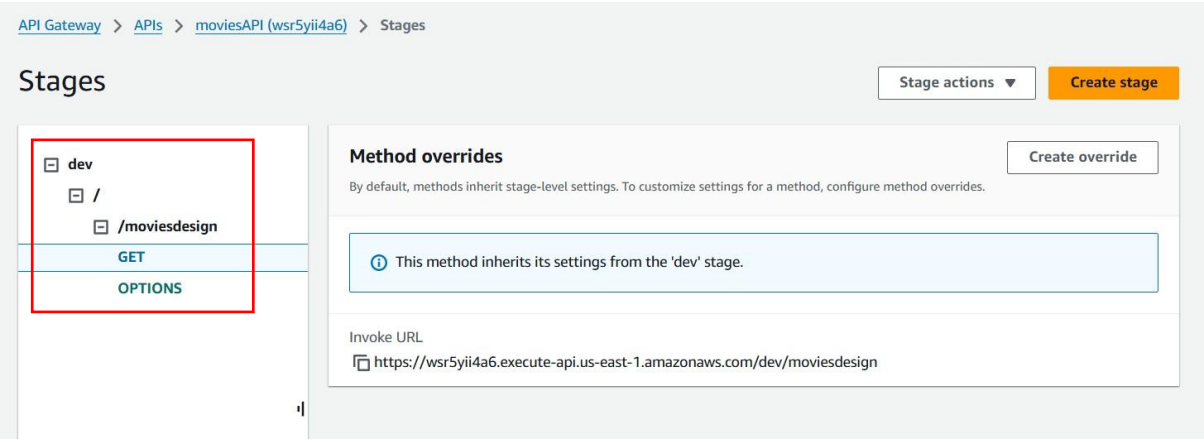
Once it is deployed now, we are going to test it out, so go to the GET method and this time choose Test and, in the query, strings give the same query as you can see below. Then click on test.



Here you can see that we get the response with status code 200 which means that it succeeded. Also, if you copy the URL and paste it in a new tab a file will be downloaded. Open this file in paint you will see an image of a cat.



Now in your API Gateway from the left pane choose stages. Then you need to expand your dev stage and go to GET method then copy the invoke URL.



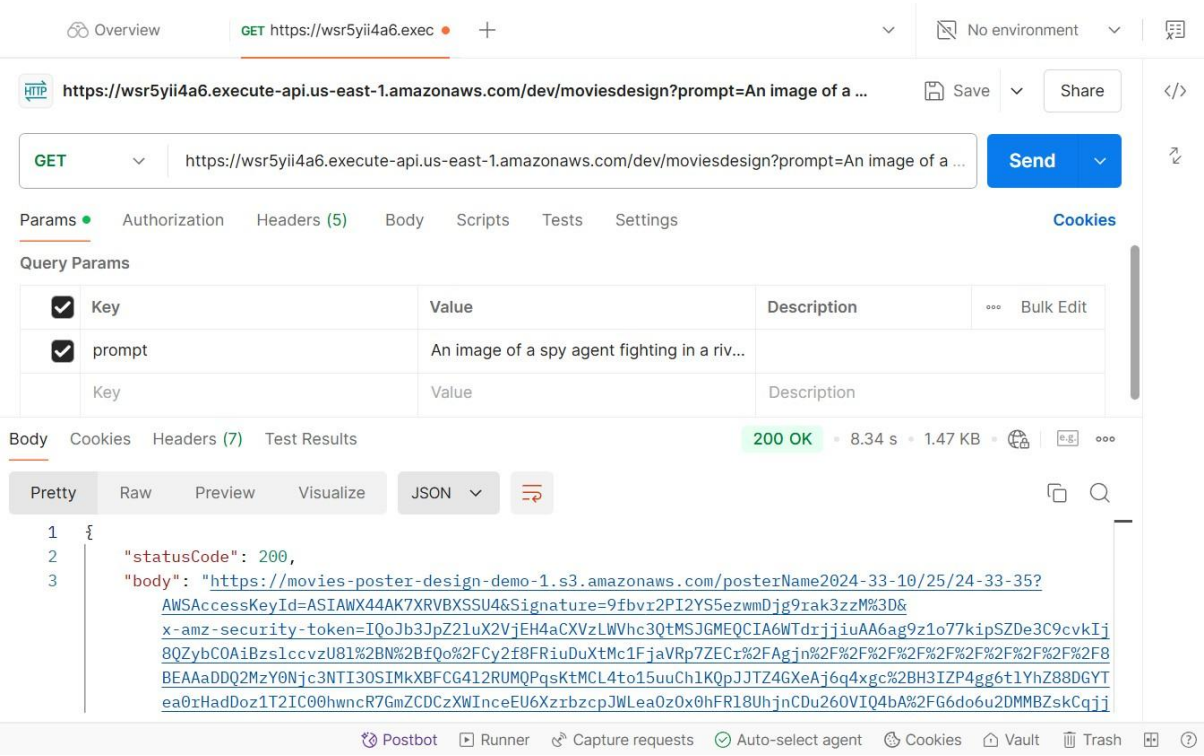
In your laptop you need to open the Post Man tool. In the tool create a workspace and choose GET method, then paste the invoke URL you copied from API gateway here.

After that in the key write prompt and, in the prompt, you need write this sentence given below. Click on Send.

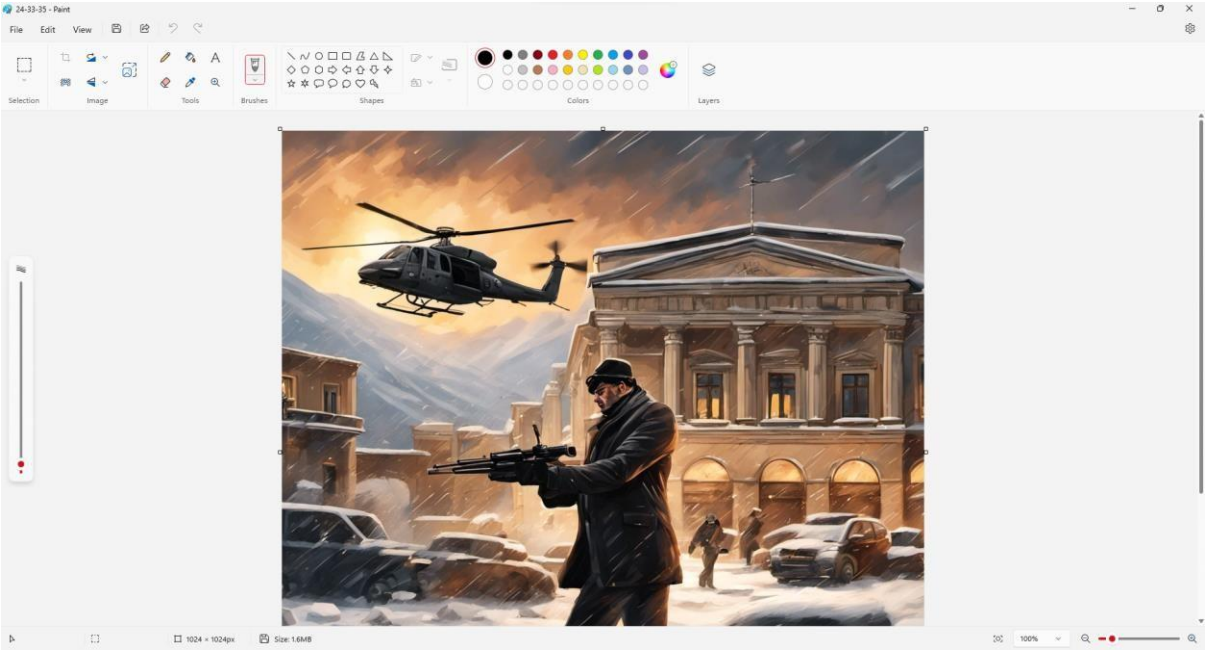
Then you will get the status code 200 and the URL. Copy the URL and paste it in the browser then a file will be downloaded.

Open that file in Paint app in your laptop.

"An image of a spy agent fighting in a rival country with guns and helicopters with backdrop of a shopping complex with heavy snow and old Greek architecture building late in the evening with sunsetting behind mountains. The iamge should be a photograph with Aaron Jasinski style"



In the paint app you will see the image.


24-33-35 - Paint
File
Edit
View
Selection
Image
Tools
Brushes
Shapes
Colors
Layers
1024 × 1024px
Size: 1.6MB
100%